PINCH MOVES, DEFINITE LATTICES AND THE NONORIENTABLE FOUR-GENUS OF KNOTS WITH AT MOST THIRTEEN CROSSINGS

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ABSTRACT. The smooth nonorientable four-genus $\gamma_4(K)$ of a knot is the least first Betti number of a smooth nonorientable surface in B^4 bounded by K. KnotInfo leaves it undetermined for 3525 prime knots with eleven to thirteen crossings. We determine it for 2537 of them and narrow the interval for 376 more. Upper bounds come from pinch moves, nonorientable band moves at a crossing of a diagram that turn the knot into a slice knot or into a knot whose nonorientable four-genus is known; each is a certificate that a program independent of the search checks by Reidemeister moves or by an isomorphism of triangulated knot complements. Lower bounds come from Donaldson's diagonalisation theorem through the obstructions of Jabuka and Kelly, which we evaluate on the definite Goeritz lattices of the alternating knots by an exhaustive search for embeddings into the standard lattice.

1. INTRODUCTION

Every knot $K \subset S^3 = \partial B^4$ bounds a smooth surface in B^4; if K is not slice, it bounds nonorientable ones as well. The *nonorientable four-genus* $\gamma_4(K)$ is the least first Betti number of a smooth compact nonorientable surface $F \subset B^4$ with $\partial F = K$; for slice knots one sets $\gamma_4(K) = 1$. It was studied by Murakami and Yasuhara [11] and by Gilmer and Livingston [5]. Its values on small knots were computed by Jabuka and Kelly for eight and nine crossings [8], by Ghanbarian for ten [4] and by Fairchild for nonalternating knots with eleven [3]. Lee and Sabloff [10] extended the computation to fourteen crossings with an algorithmic search for Möbius bands in the style of Dunfield and Gong [1], together with the Arf invariant obstruction of Yasuhara [14] and a linking form obstruction of Murakami and Yasuhara [11]. The KnotInfo database [9] incorporates these data. In its version 2026.10.5, 3525 knots with eleven to thirteen crossings have an interval $[a, b]$ with $a < b$ as the entry for γ_4.

Theorem 1. *The value $\gamma_4(K)$ is determined for 2537 of these knots: 520 have $\gamma_4(K) = 1$, 1995 have $\gamma_4(K) = 2$ and 22 have $\gamma_4(K) = 3$. For 2177 of them a pinch move brings the upper end of the KnotInfo interval down to its lower end; for 163 the lower bound of Theorem 4 raises the lower end to the upper end; for 197 the two meet in between. For 376 further knots the interval is narrowed.*

Table 1 gives the counts by crossing number and by KnotInfo interval. The upper bounds use one elementary move.

Definition 2. Let D be a diagram of a knot K and c a crossing of D. Of the two smoothings of c, one respects the orientation of K and gives a two-component link; the other gives a knot K'. We say K' is obtained from K by a *pinch move* at c.

Lemma 3. *If K' is obtained from K by a pinch move, then $\gamma_4(K) \leq 1$ when K' is slice, and $\gamma_4(K) \leq \gamma_4(K') + 1$ in general.*

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Proof. The pinch move is a band move, so there is a cobordism $\Sigma \subset S^3 \times [0,1]$ from K to K' formed by $K \times [0,1]$ and one band. Since K' is connected, the band does not respect the orientation of K, and Σ is a Möbius band with an open disk removed. If K' bounds a disk Δ in B^4, then $\Sigma \cup \Delta$ is a Möbius band bounded by K, so $\gamma_4(K) \leq 1$. If K' bounds a surface F with $b_1(F) = \gamma_4(K')$, then $\Sigma \cup F$ is a nonorientable surface bounded by K with $b_1 = b_1(F) + 1$. This is the bound used by Jabuka and Kelly [8]. □

A knot that is not slice has $\gamma_4 \geq 1$. For the knots whose KnotInfo interval starts at 2, the lower bound is the obstruction to bounding a Möbius band recorded by KnotInfo from Lee and Sabloff [10]: either Yasuhara's condition $4\,\mathrm{Arf}(K) - \sigma(K) \in \{0, \pm 2\} \pmod 8$ fails [14], or the linking form obstruction of Murakami and Yasuhara [11] applies. The further lower bounds are those of Section 3. Theorem 1 therefore rests on these obstructions, on the slice status and four-genus values recorded in KnotInfo for the targets, on the certificates of Section 2 and on the lattice computations of Section 3.

2. Certificates and their verification

A certificate for a knot K consists of a planar diagram code D, a crossing c of D, a choice of smoothing, and a target: either the unknot or a knot T of KnotInfo. It asserts that

(1) D is a diagram of K;
(2) the chosen smoothing of c gives a one-component diagram D';
(3) D' represents T or its mirror image;
(4) the bound $\beta = 1$ if T is slice or the unknot, and $\beta = \gamma_4(T) + 1$ otherwise, is smaller than the upper end of the KnotInfo interval of K.

Since γ_4 and sliceness are invariant under mirror image, (1) to (4) and Lemma 3 give $\gamma_4(K) \leq \beta$.

The certificates were found with SnapPy [13]: for every open knot we applied every pinch move to the KnotInfo diagram and to up to twenty-five diagrams obtained from it by random Reidemeister moves, and identified the resulting knots by hyperbolic structure. A candidate enters the file only if the verifier below accepts it; when it did not, the search went on with other crossings and other diagrams of the same knot. A second pass used the knots settled in the first pass as targets; the number of certificates with such a target is 2, and the verifier checks them after the certificates of their targets. A last pass applied every pinch move to twenty diagrams of each knot left open.

The verifier `verify.py` does not use SnapPy. It reads the KnotInfo database and uses Regina [12] for the topology. For (1) and (3) it proves that two diagrams represent the same knot up to mirror image in one of two ways.

- *Isotopy.* Both diagrams are simplified by Reidemeister moves to the same number of crossings, and the closures of the two minimal diagrams under Reidemeister moves that add at most one crossing share a diagram. Diagrams are compared by Regina's canonical signatures, taken up to reflection, rotation and reversal.
- *Complements.* The triangulated complements of the two diagrams have combinatorially isomorphic triangulations, possibly after Pachner moves that add at most one tetrahedron, or one of the triangulations obtained from forty simplifications of each complement under fixed seeds of Regina's random engine, together with the triangulations one Pachner move away from the smallest of them, occurs for both diagrams. Each of these triangulations is a triangulation of the complement it came from, so a common one shows that the complements are homeomorphic, and by the theorem of Gordon and Luecke [7] the knots agree up to mirror image.

Both tests are exact: a positive answer is a proof. Regina's simplification is randomised, so the verifier makes up to four attempts, each from a fixed seed, and gives each certificate two minutes. For the unknot the verifier uses Regina's recognition of the solid torus. Item (2) is checked by following the strands of D', and (4) by reading KnotInfo.

The verifier accepts all 2728 certificates. It rejects certificates corrupted in four ways, on a sample of twenty: the other smoothing of the crossing, the next crossing, a wrong target, and a bound lowered by one (Table 2). The seven modified certificates with the next crossing that it accepts are valid certificates: that crossing gives the same target.

3. Lower bounds from definite lattices

Let D be a diagram of a knot K with a checkerboard colouring. Following Gordon and Litherland [6], give each crossing c the weight $\eta(c) = \pm 1$ according to the position of the shaded regions at c, let $X_0, \dots, X_n$ be the unshaded regions, and let G be the $n \times n$ matrix with $g_{ij} = -\sum \eta(c)$ over the crossings meeting X_i and X_j for $i \neq j$, and $g_{ii} = -\sum_{k \neq i} g_{ik}$ with k running from 0 to n. This is the Goeritz matrix; $\det G = \pm \det K$, and with $\mu = \sum \eta(c)$ over the crossings of type II for the shaded surface,

$$\sigma(K) = \operatorname{sign}(G) - \mu. \tag{1}$$

The form G is the intersection form of the double cover of B^4 branched along the shaded surface pushed into B^4 [6]. If D is reduced and alternating, the forms G_+ and G_- of the two colourings are positive and negative definite.

Theorem 4 (Jabuka and Kelly [8, Theorems 2.10–2.12])**.** *Let K be a knot with a reduced alternating diagram, let $P_+ = G_+$ and $P_- = -G_-$, of ranks $r_\pm$, and let $s \equiv \sigma(K) + 4\operatorname{Arf}(K)$ (mod 8). Call $\ell \in \mathbb{N}$ admissible if $\ell \mid \det K$ and $\det K/\ell$ is a square.*

1. *If $s = 2$ and $P_- \oplus [\ell]$ embeds in $\mathbb{Z}^{r_- + 1}$ for no admissible ℓ, or $s = 6$ and $P_+ \oplus [\ell]$ embeds in $\mathbb{Z}^{r_+ + 1}$ for no admissible ℓ, then $\gamma_4(K) \geq 2$.*
2. *If $s = 0$ and neither $P_+ \oplus [\ell]$ embeds in $\mathbb{Z}^{r_+ + 1}$ nor $P_- \oplus [\ell]$ in $\mathbb{Z}^{r_- + 1}$, for any admissible ℓ, then $\gamma_4(K) \geq 2$.*
3. *If $s = 4$ and neither P_+ embeds in $\mathbb{Z}^{r_+ + 2}$ nor P_- in $\mathbb{Z}^{r_- + 2}$, then $\gamma_4(K) \geq 3$.*

Here $\mathbb{Z}^N$ carries the standard positive definite form.

We recall the argument. If K bounds a Möbius band $\Sigma \subset B^4$, the double branched cover $W(\Sigma)$ has $b_1 = 0$ [8] and $b_2 = 1$, its form is $[\pm\ell]$ with ℓ admissible by a theorem of Gilmer and of Owens and Strle [8, Corollary 2.8], and the congruence of Gilmer and Livingston [5] fixes the sign of ℓ by s. Gluing $W(\Sigma)$ with reversed orientation to the double cover bounded by the shaded surface whose Goeritz form has the opposite sign gives a closed definite smooth four-manifold whose form contains $G \oplus [\mp\ell]$ in full rank. By Donaldson's theorem [2] that form is diagonal; changing the sign of the whole form gives (1) and (2). In (3) a surface with $b_1 = 2$ would give a definite $W(\Sigma)$ of rank two, of either sign, and the same gluing embeds P_+ or P_- in a diagonal lattice of rank two more. In (1) the statement of [8] names the target lattice as ε-definite; the argument above, which is theirs, produces the lattice of the sign of G, and it is in this form that we apply it.

One convention needs a word. Since γ_4 is invariant under mirror image, and mirroring changes the sign of every weight $\eta(c)$, of G, of μ and of σ, it does not matter which of the two weight conventions is called positive, as long as σ is computed from (1) with the same weights.

Lemma 5. *Let P be a positive definite lattice of rank r and $\ell \in \mathbb{N}$. Then $P \oplus [\ell]$ embeds in $\mathbb{Z}^{r+1}$ if and only if there are an embedding $\phi\colon P \to \mathbb{Z}^{r+1}$ and an integer $k \geq 1$ with $\ell = k^2|w|^2$, where w is a primitive vector spanning the orthogonal complement of $\phi(P)$.*

Proof. The orthogonal complement of a sublattice of rank r in $\mathbb{Z}^{r+1}$ is $\mathbb{Z}w$ for a primitive w, and a vector of norm ℓ orthogonal to $\phi(P)$ is kw with $k^2|w|^2 = \ell$. □

Lemma 6. *Let P be a positive definite Gram matrix of rank r and $N \geq r$. Every embedding of P in $\mathbb{Z}^N$ is carried by a signed permutation of the coordinates to one found by the following search. The basis vectors $e_1, \dots, e_r$ are placed in turn; if the vectors already placed are supported on the first t coordinates, the next vector is $(x, y, 0)$ with $x \in \mathbb{Z}^t$ arbitrary, subject to the norm and the inner products with the vectors already placed, and y a nonincreasing sequence of positive integers.*

Proof. By induction on the number of vectors placed. Given an embedding $v_1, \dots, v_r$ whose first $i-1$ vectors are of the stated form, supported on the first t coordinates, the signed permutations of the coordinates beyond t fix $v_1, \dots, v_{i-1}$; one of them makes the entries of v_i beyond t nonnegative and nonincreasing, with its nonzero entries first, and it does not change the shape of the vectors already placed. □

We take the KnotInfo diagram of each open alternating knot, compute both Goeritz forms and σ from (1), check that the two colourings give the same σ, that $|\sigma|$ agrees with KnotInfo and that the forms are definite, and run the search of Lemma 6 to the end: for (1) and (2) it lists the norms $|w|^2$ of Lemma 5 over all embeddings, for (3) it decides whether an embedding exists. Of the 3525 knots, the hypotheses of Theorem 4 hold and raise the KnotInfo lower end for 382: 360 reach $\gamma_4 \geq 2$ (314 by (1) and 46 by (2)) and 22 reach $\gamma_4 \geq 3$ by (3). The data are in `lower_bounds.json`.

The program `verify_lower.py` recomputes every entry with the basis order of the search shuffled under a fixed seed, and runs three tests of the conventions. On the alternating knots with at most eleven crossings whose γ_4 KnotInfo records, the bound never exceeds the recorded value, while the same program with the sign of σ reversed contradicts the recorded value for more than two hundred of them. On the alternating knots of `certificates.json`, the bound never exceeds the upper bound of the pinch certificate.

4. Results

The remaining 988 knots, with their narrowed intervals, are listed in `still_open.json`. For a knot with interval $[1, 2]$ that is not settled, no pinch move we tried gives a slice knot, and Theorem 4 does not apply or does not obstruct; this says nothing about whether $\gamma_4 = 1$. Theorem 4 needs a definite diagram, so it says nothing about the nonalternating knots.

Remark 7. The 520 Möbius bands found here are not in the data of Lee and Sabloff [10], whose search follows Dunfield and Gong [1] and runs over a finite family of bands chosen in advance. We have not determined which restriction of that family excludes them. The knots settled at $\gamma_4 = 2$ were outside their aim, which was Möbius bands; for them the target of the pinch move is a knot known to bound a Möbius band but not a disk.

Appendix A. The settled knots

$\gamma_4 = 1$ **(520 knots).**

12a_133	12a_390	12a_750	12a_1069	12a_1208	12n_495
12a_200	12a_457	12a_818	12a_1070	12n_32	12n_554
12a_212	12a_474	12a_878	12a_1092	12n_70	12n_555
12a_218	12a_611	12a_882	12a_1094	12n_147	12n_597
12a_313	12a_639	12a_895	12a_1102	12n_382	12n_632
12a_335	12a_698	12a_916	12a_1115	12n_390	12n_642
12a_373	12a_709	12a_932	12a_1206	12n_444	12n_769

crossings	KnotInfo	knots	settled	narrowed	open
11	[1, 2]	14	8	0	6
11	[1, 3]	9	8	1	0
11	[2, 3]	36	33	0	3
12	[1, 2]	87	43	0	44
12	[1, 3]	87	49	35	3
12	[2, 3]	278	265	0	13
12	[2, 4]	3	3	0	0
13	[1, 2]	890	431	0	459
13	[1, 3]	677	318	338	21
13	[1, 4]	3	1	2	0
13	[1, 5]	1	0	0	1
13	[2, 3]	1437	1375	0	62
13	[2, 4]	3	3	0	0
all		3525	2537	376	612

TABLE 1. Open entries of γ_4 in KnotInfo 2026.10.5 and what the certificates give. "Settled": the value is now known; "lowered": the upper bound is lower.

corruption	rejected
other smoothing of the crossing	20 of 20
next crossing of the diagram	13 of 20
wrong target ($12a_1$)	20 of 20
bound lowered by one	20 of 20

TABLE 2. Corrupted certificates rejected by the verifier.

12n_784	13a_307	13a_619	13a_988	13a_1250	13a_1627
12n_831	13a_318	13a_680	13a_990	13a_1277	13a_1648
12n_843	13a_338	13a_687	13a_995	13a_1316	13a_1656
13a_5	13a_342	13a_699	13a_1000	13a_1320	13a_1658
13a_6	13a_343	13a_712	13a_1027	13a_1324	13a_1665
13a_12	13a_353	13a_714	13a_1068	13a_1362	13a_1666
13a_45	13a_365	13a_721	13a_1072	13a_1380	13a_1671
13a_79	13a_369	13a_726	13a_1077	13a_1412	13a_1673
13a_129	13a_375	13a_730	13a_1087	13a_1417	13a_1712
13a_131	13a_381	13a_732	13a_1103	13a_1421	13a_1719
13a_136	13a_392	13a_734	13a_1106	13a_1436	13a_1722
13a_138	13a_396	13a_776	13a_1122	13a_1438	13a_1727
13a_156	13a_410	13a_777	13a_1131	13a_1454	13a_1731
13a_158	13a_476	13a_786	13a_1134	13a_1477	13a_1756
13a_166	13a_505	13a_815	13a_1140	13a_1506	13a_1758
13a_180	13a_518	13a_823	13a_1163	13a_1511	13a_1767
13a_197	13a_519	13a_824	13a_1169	13a_1514	13a_1851
13a_224	13a_530	13a_827	13a_1176	13a_1523	13a_1856
13a_232	13a_537	13a_845	13a_1186	13a_1532	13a_1862
13a_244	13a_538	13a_861	13a_1193	13a_1598	13a_1907
13a_247	13a_550	13a_906	13a_1230	13a_1608	13a_2059
13a_253	13a_591	13a_916	13a_1231	13a_1609	13a_2094
13a_265	13a_610	13a_925	13a_1237	13a_1624	13a_2166

13a_2218	13a_3408	13a_4289	13n_406	13n_2080	13n_3902
13a_2308	13a_3413	13a_4338	13n_418	13n_2088	13n_3916
13a_2311	13a_3415	13a_4342	13n_428	13n_2115	13n_3919
13a_2323	13a_3417	13a_4365	13n_520	13n_2150	13n_3921
13a_2328	13a_3418	13a_4367	13n_529	13n_2193	13n_3923
13a_2339	13a_3462	13a_4394	13n_530	13n_2245	13n_3933
13a_2389	13a_3463	13a_4398	13n_561	13n_2265	13n_3968
13a_2402	13a_3475	13a_4403	13n_563	13n_2383	13n_4023
13a_2411	13a_3477	13a_4412	13n_620	13n_2404	13n_4030
13a_2414	13a_3491	13a_4419	13n_635	13n_2406	13n_4107
13a_2502	13a_3494	13a_4426	13n_671	13n_2409	13n_4115
13a_2505	13a_3497	13a_4430	13n_813	13n_2453	13n_4152
13a_2506	13a_3505	13a_4437	13n_816	13n_2503	13n_4155
13a_2511	13a_3515	13a_4443	13n_839	13n_2610	13n_4162
13a_2560	13a_3522	13a_4449	13n_852	13n_2629	13n_4193
13a_2595	13a_3556	13a_4475	13n_855	13n_2690	13n_4202
13a_2632	13a_3579	13a_4479	13n_892	13n_2694	13n_4247
13a_2645	13a_3598	13a_4486	13n_932	13n_2706	13n_4272
13a_2676	13a_3609	13a_4487	13n_988	13n_2741	13n_4300
13a_2788	13a_3611	13a_4493	13n_994	13n_2765	13n_4305
13a_2828	13a_3618	13a_4511	13n_999	13n_2773	13n_4306
13a_2864	13a_3653	13a_4513	13n_1005	13n_2812	13n_4313
13a_2870	13a_3657	13a_4524	13n_1057	13n_2834	13n_4333
13a_2893	13a_3674	13a_4544	13n_1061	13n_2837	13n_4334
13a_2925	13a_3689	13a_4577	13n_1062	13n_2851	13n_4351
13a_3012	13a_3692	13a_4591	13n_1067	13n_2857	13n_4354
13a_3026	13a_3741	13a_4603	13n_1111	13n_2926	13n_4380
13a_3028	13a_3758	13a_4620	13n_1114	13n_2942	13n_4457
13a_3039	13a_3764	13a_4629	13n_1201	13n_2949	13n_4477
13a_3046	13a_3772	13a_4647	13n_1243	13n_2966	13n_4485
13a_3057	13a_3776	13a_4648	13n_1275	13n_2977	13n_4509
13a_3058	13a_3786	13a_4654	13n_1277	13n_2985	13n_4515
13a_3076	13a_3787	13a_4696	13n_1302	13n_3108	13n_4517
13a_3167	13a_3799	13a_4697	13n_1328	13n_3190	13n_4521
13a_3181	13a_3805	13a_4705	13n_1419	13n_3198	13n_4543
13a_3188	13a_3807	13a_4743	13n_1490	13n_3236	13n_4614
13a_3190	13a_3811	13a_4744	13n_1492	13n_3311	13n_4632
13a_3198	13a_3866	13a_4746	13n_1494	13n_3334	13n_4642
13a_3242	13a_3905	13a_4794	13n_1500	13n_3338	13n_4657
13a_3258	13a_3928	13a_4848	13n_1582	13n_3344	13n_4666
13a_3266	13a_3930	13a_4874	13n_1585	13n_3351	13n_4709
13a_3274	13a_3955	13n_7	13n_1609	13n_3402	13n_4746
13a_3275	13a_3993	13n_30	13n_1628	13n_3452	13n_4870
13a_3279	13a_4054	13n_50	13n_1638	13n_3501	13n_4896
13a_3294	13a_4077	13n_54	13n_1670	13n_3526	13n_4940
13a_3298	13a_4103	13n_78	13n_1677	13n_3602	13n_5006
13a_3302	13a_4119	13n_149	13n_1678	13n_3641	13n_5009
13a_3306	13a_4141	13n_153	13n_1688	13n_3667	13n_5012
13a_3328	13a_4152	13n_157	13n_1693	13n_3695	13n_5015
13a_3333	13a_4178	13n_231	13n_1726	13n_3723	13n_5041
13a_3343	13a_4195	13n_290	13n_1789	13n_3739	13n_5067
13a_3354	13a_4198	13n_330	13n_1932	13n_3740	13n_5089
13a_3355	13a_4211	13n_341	13n_1935	13n_3745	13n_5095
13a_3356	13a_4220	13n_344	13n_1976	13n_3752	13n_5104
13a_3374	13a_4247	13n_347	13n_1992	13n_3771	13n_5108
13a_3399	13a_4249	13n_360	13n_2021	13n_3776	
13a_3403	13a_4268	13n_370	13n_2022	13n_3899	

$\gamma_4 = 2$ **(1995 knots).**

11a_6	12a_44	12a_460	12a_762	12a_1006	12a_1248
11a_17	12a_47	12a_461	12a_763	12a_1009	12a_1252
11a_20	12a_49	12a_480	12a_767	12a_1012	12a_1255
11a_29	12a_52	12a_493	12a_783	12a_1013	12a_1256
11a_49	12a_55	12a_502	12a_787	12a_1015	12a_1260
11a_51	12a_64	12a_503	12a_792	12a_1021	12a_1265
11a_53	12a_66	12a_507	12a_795	12a_1025	12a_1266
11a_59	12a_67	12a_508	12a_796	12a_1030	12a_1270
11a_64	12a_76	12a_514	12a_801	12a_1031	12a_1274
11a_66	12a_85	12a_523	12a_802	12a_1035	12a_1276
11a_89	12a_88	12a_533	12a_806	12a_1036	12a_1279
11a_144	12a_96	12a_542	12a_811	12a_1037	12a_1288
11a_161	12a_97	12a_546	12a_812	12a_1044	12n_27
11a_163	12a_103	12a_551	12a_813	12a_1049	12n_33
11a_173	12a_113	12a_553	12a_815	12a_1052	12n_52
11a_178	12a_114	12a_556	12a_816	12a_1058	12n_67
11a_188	12a_117	12a_570	12a_817	12a_1059	12n_104
11a_192	12a_134	12a_575	12a_820	12a_1079	12n_107
11a_203	12a_136	12a_587	12a_822	12a_1080	12n_148
11a_206	12a_137	12a_590	12a_827	12a_1084	12n_153
11a_214	12a_143	12a_592	12a_828	12a_1088	12n_164
11a_240	12a_146	12a_595	12a_829	12a_1091	12n_169
11a_241	12a_148	12a_597	12a_832	12a_1097	12n_173
11a_248	12a_152	12a_610	12a_840	12a_1099	12n_174
11a_259	12a_156	12a_616	12a_841	12a_1103	12n_185
11a_264	12a_167	12a_628	12a_844	12a_1106	12n_186
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13a_4673	13a_4862	13n_550	13n_1077	13n_1594	13n_2372
13a_4691	13a_4864	13n_569	13n_1080	13n_1595	13n_2421
13a_4695	13a_4865	13n_596	13n_1086	13n_1643	13n_2426
13a_4698	13a_4869	13n_601	13n_1095	13n_1654	13n_2437
13a_4704	13a_4873	13n_616	13n_1097	13n_1656	13n_2442
13a_4706	13a_4875	13n_618	13n_1104	13n_1660	13n_2487
13a_4707	13a_4876	13n_622	13n_1106	13n_1679	13n_2515
13a_4709	13n_6	13n_629	13n_1115	13n_1681	13n_2530
13a_4718	13n_17	13n_631	13n_1117	13n_1684	13n_2536
13a_4725	13n_25	13n_645	13n_1176	13n_1703	13n_2561
13a_4732	13n_32	13n_663	13n_1180	13n_1730	13n_2572
13a_4733	13n_41	13n_670	13n_1187	13n_1732	13n_2586
13a_4737	13n_44	13n_675	13n_1192	13n_1771	13n_2603
13a_4739	13n_45	13n_679	13n_1204	13n_1784	13n_2608
13a_4740	13n_75	13n_680	13n_1205	13n_1791	13n_2649
13a_4745	13n_82	13n_681	13n_1212	13n_1838	13n_2657
13a_4747	13n_113	13n_691	13n_1217	13n_1851	13n_2661
13a_4757	13n_131	13n_709	13n_1222	13n_1858	13n_2664
13a_4758	13n_150	13n_719	13n_1225	13n_1892	13n_2672
13a_4764	13n_154	13n_723	13n_1229	13n_1906	13n_2677
13a_4766	13n_163	13n_732	13n_1235	13n_1907	13n_2689
13a_4767	13n_177	13n_744	13n_1238	13n_1927	13n_2711
13a_4768	13n_186	13n_747	13n_1239	13n_1931	13n_2719
13a_4770	13n_187	13n_755	13n_1240	13n_1933	13n_2722
13a_4771	13n_204	13n_767	13n_1273	13n_1936	13n_2751
13a_4774	13n_215	13n_771	13n_1278	13n_1950	13n_2775
13a_4787	13n_222	13n_773	13n_1294	13n_1998	13n_2805
13a_4790	13n_225	13n_776	13n_1300	13n_2042	13n_2816
13a_4792	13n_241	13n_787	13n_1323	13n_2061	13n_2827

13n_2870	13n_3305	13n_3735	13n_4186	13n_4576	13n_4879
13n_2881	13n_3333	13n_3736	13n_4187	13n_4588	13n_4884
13n_2883	13n_3337	13n_3743	13n_4189	13n_4589	13n_4887
13n_2898	13n_3354	13n_3749	13n_4200	13n_4599	13n_4890
13n_2899	13n_3368	13n_3774	13n_4207	13n_4605	13n_4891
13n_2900	13n_3419	13n_3780	13n_4208	13n_4615	13n_4899
13n_2901	13n_3426	13n_3785	13n_4218	13n_4622	13n_4900
13n_2907	13n_3427	13n_3788	13n_4221	13n_4633	13n_4906
13n_2920	13n_3432	13n_3799	13n_4223	13n_4637	13n_4911
13n_2929	13n_3435	13n_3801	13n_4243	13n_4641	13n_4914
13n_2935	13n_3447	13n_3806	13n_4256	13n_4644	13n_4928
13n_2939	13n_3454	13n_3807	13n_4261	13n_4649	13n_4931
13n_2959	13n_3461	13n_3810	13n_4263	13n_4650	13n_4933
13n_2972	13n_3470	13n_3813	13n_4269	13n_4656	13n_4936
13n_2981	13n_3476	13n_3822	13n_4278	13n_4660	13n_4941
13n_2995	13n_3484	13n_3824	13n_4289	13n_4673	13n_4943
13n_3000	13n_3486	13n_3836	13n_4314	13n_4676	13n_4949
13n_3003	13n_3491	13n_3837	13n_4341	13n_4679	13n_4951
13n_3010	13n_3493	13n_3849	13n_4342	13n_4687	13n_4952
13n_3015	13n_3497	13n_3852	13n_4343	13n_4693	13n_4960
13n_3034	13n_3499	13n_3854	13n_4353	13n_4700	13n_4966
13n_3038	13n_3502	13n_3873	13n_4359	13n_4704	13n_4967
13n_3046	13n_3508	13n_3879	13n_4360	13n_4714	13n_4972
13n_3048	13n_3515	13n_3882	13n_4361	13n_4717	13n_4975
13n_3051	13n_3518	13n_3907	13n_4362	13n_4729	13n_4978
13n_3060	13n_3520	13n_3913	13n_4363	13n_4730	13n_4981
13n_3063	13n_3532	13n_3920	13n_4365	13n_4731	13n_4982
13n_3068	13n_3566	13n_3928	13n_4366	13n_4739	13n_4985
13n_3083	13n_3576	13n_3942	13n_4373	13n_4743	13n_4990
13n_3103	13n_3579	13n_3990	13n_4377	13n_4747	13n_4991
13n_3111	13n_3583	13n_4008	13n_4402	13n_4751	13n_4993
13n_3113	13n_3589	13n_4015	13n_4404	13n_4752	13n_4994
13n_3124	13n_3593	13n_4029	13n_4408	13n_4761	13n_4997
13n_3137	13n_3595	13n_4043	13n_4410	13n_4767	13n_5003
13n_3145	13n_3608	13n_4050	13n_4418	13n_4774	13n_5013
13n_3156	13n_3633	13n_4054	13n_4447	13n_4782	13n_5014
13n_3188	13n_3643	13n_4057	13n_4455	13n_4791	13n_5021
13n_3199	13n_3648	13n_4073	13n_4463	13n_4793	13n_5022
13n_3200	13n_3653	13n_4085	13n_4469	13n_4795	13n_5030
13n_3210	13n_3657	13n_4094	13n_4501	13n_4804	13n_5044
13n_3212	13n_3673	13n_4095	13n_4520	13n_4813	13n_5046
13n_3216	13n_3676	13n_4129	13n_4537	13n_4820	13n_5054
13n_3239	13n_3697	13n_4133	13n_4546	13n_4822	13n_5070
13n_3248	13n_3698	13n_4137	13n_4550	13n_4837	13n_5075
13n_3254	13n_3709	13n_4164	13n_4556	13n_4846	13n_5084
13n_3258	13n_3712	13n_4165	13n_4561	13n_4847	13n_5100
13n_3274	13n_3714	13n_4169	13n_4565	13n_4853	13n_5102
13n_3276	13n_3721	13n_4179	13n_4567	13n_4865	
13n_3293	13n_3724	13n_4181	13n_4568	13n_4867	
13n_3294	13n_3729	13n_4183	13n_4569	13n_4868	

$\gamma_4 = 3$ **(22 knots).**

12a_561	12a_906	13a_1557	13a_1992	13a_3585	13a_4637
12a_569	12a_907	13a_1662	13a_2890	13a_3747	13a_4674
12a_664	12a_1123	13a_1826	13a_3034	13a_3936	
12a_780	13a_1398	13a_1974	13a_3121	13a_4566	

Data availability

The certificates, the lower bounds, both verifiers and the search programs are in the folder `crosscap4d` of the repository `https://github.com/creelie/conjecture-to-choose`, release 1.2.0, `https://github.com/creelie/conjecture-to-choose/releases/tag/v1.2.0`.

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